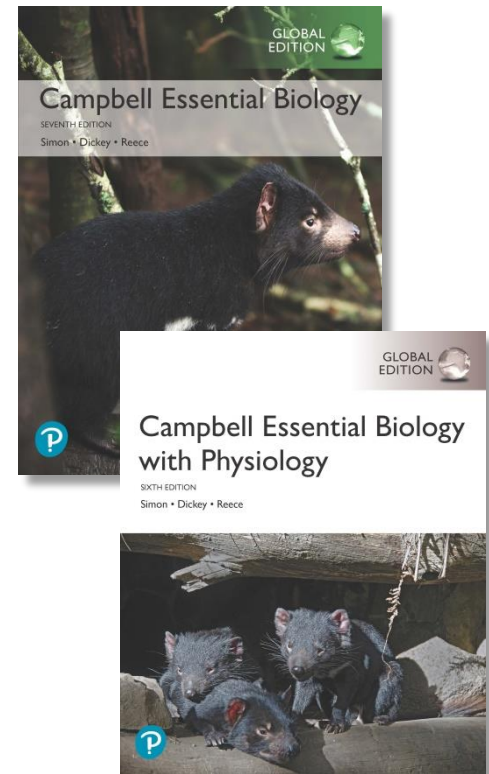


# Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

## Chapter 21 Unifying Concepts of Animal Structure and Function



PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

# It's a Good Burn! **A Moderate Fever** is not Always Your Body's Enemy ; It can **Help Your Immune System** During Battles Against Infection



# Biology and Society: An Avoidable Tragedy

- The animal body self-regulates.
  - When you play or work out hard, your muscles produce a lot of heat that in turn increases your overall internal temperature.
  - A control center in your brain initiates responses to keep your internal temperature near normal 37°C.
  - Under extreme heat exhaustion, body temperature rises to the point that it disrupts the brain's temperature control center and heat stroke occurs, causing the body's organs to begin to fail.
  - Unless the body is cooled and hydrated, brain damage and then death can follow.

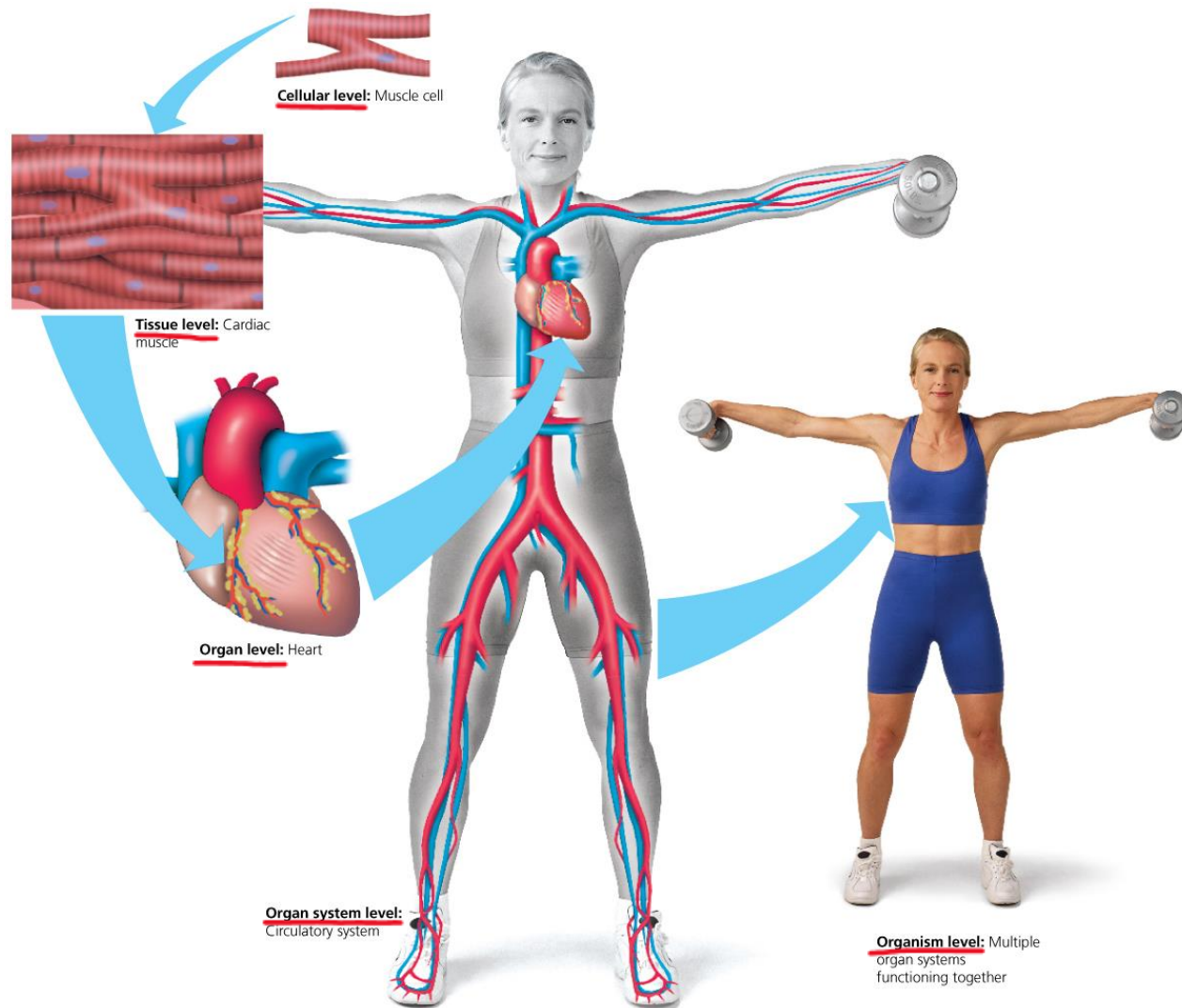
# Heat Illness



# The Structural Organization of Animals: Anatomy and Physiology

- Life is characterized by a hierarchy of organization.
- In animals,
  - individual cells are grouped into tissues,
  - tissues combine to form organs,
  - organs are organized into organ systems, and
  - organ systems make up the entire organism.
- As with all other areas of biology, a study of animal bodies will highlight the interactions that underlie life's processes.

# Structural Hierarchy in a Human



# Anatomy and Physiology

- Biologists distinguish anatomy from physiology.
  - **Anatomy** is the study of the structure of an organism's parts.
  - **Physiology** is the study of the function of those parts.
- The correlation of structure and function is a fundamental principle of biology that is evident at all levels of life's hierarchy.

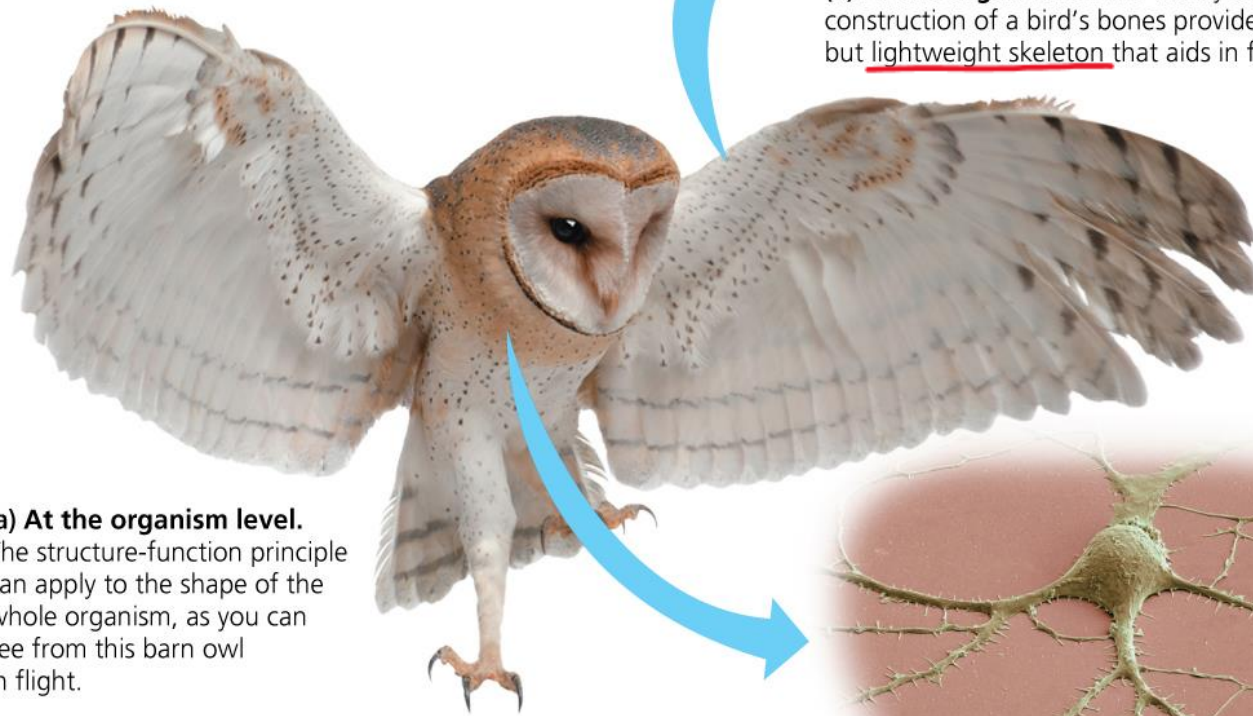


# Structure Fits Function

Honeycomb: 벌집



(b) **At the organ level.** The honeycombed construction of a bird's bones provides a strong but lightweight skeleton that aids in flight.



(a) **At the organism level.** The structure-function principle can apply to the shape of the whole organism, as you can see from this barn owl in flight.



(c) **At the cellular level.** Nerve cells that control the muscles of a bird or other animal have long extensions that transmit signals.

2,000x



# Tissues

- The cell is the basic unit of all living organisms.
- In almost all animals, including humans, cells are grouped into tissues.
  - A **tissue** is an integrated group of similar cells that performs a specific function.
  - Animals have four main categories of tissue:
    1. epithelial tissue,
    2. connective tissue,
    3. muscle tissue, and
    4. nervous tissue.

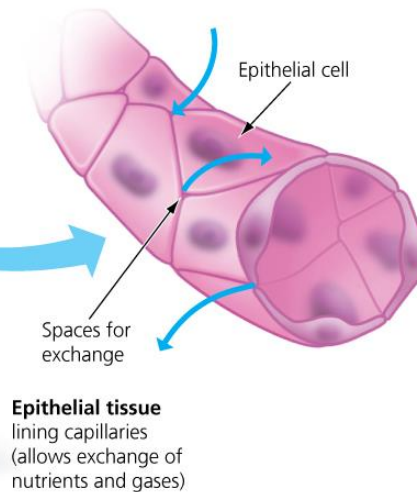
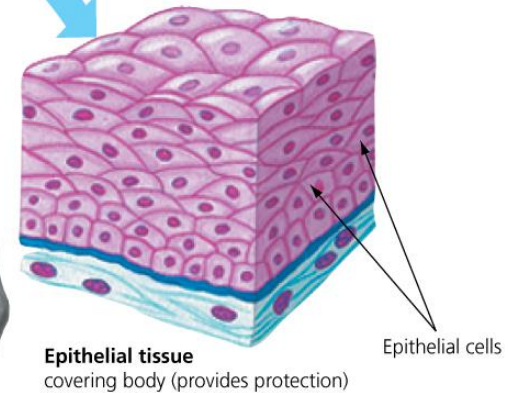
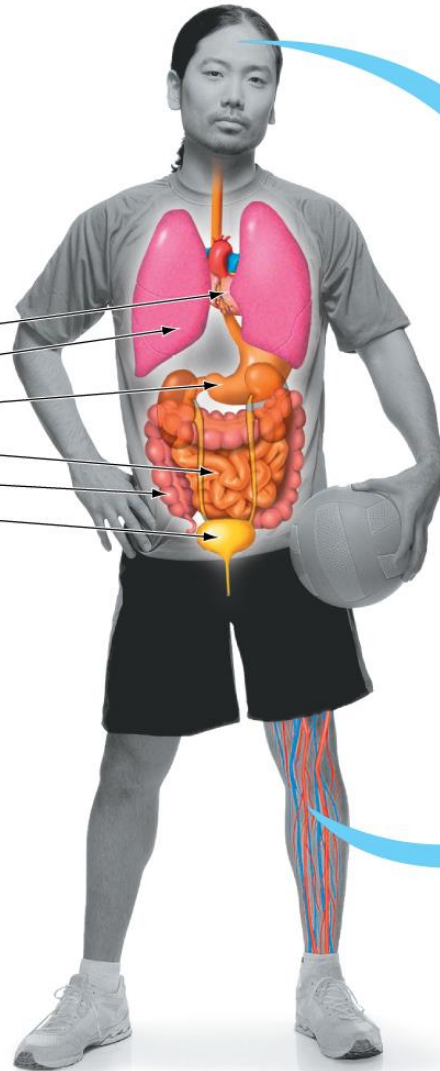
# Epithelial Tissue (1 of 2)

- **Epithelial tissue**, also known as **epithelium**, covers the surface of the body and lines organs.
- The architecture of an epithelium illustrates how structure fits function at the tissue level.
  - Your **skin** contains many layers of tightly bound epithelial cells, forming a protective, waterproof barrier that surrounds your body and keeps it safe from external threats.
  - In contrast, a single thin and leaky layer of epithelial tissue lines capillaries, where it is well suited for the role it plays of exchanging substances with the circulatory system.

# Epithelial Tissue (2 of 2)

## Examples of organs lined with epithelial tissue

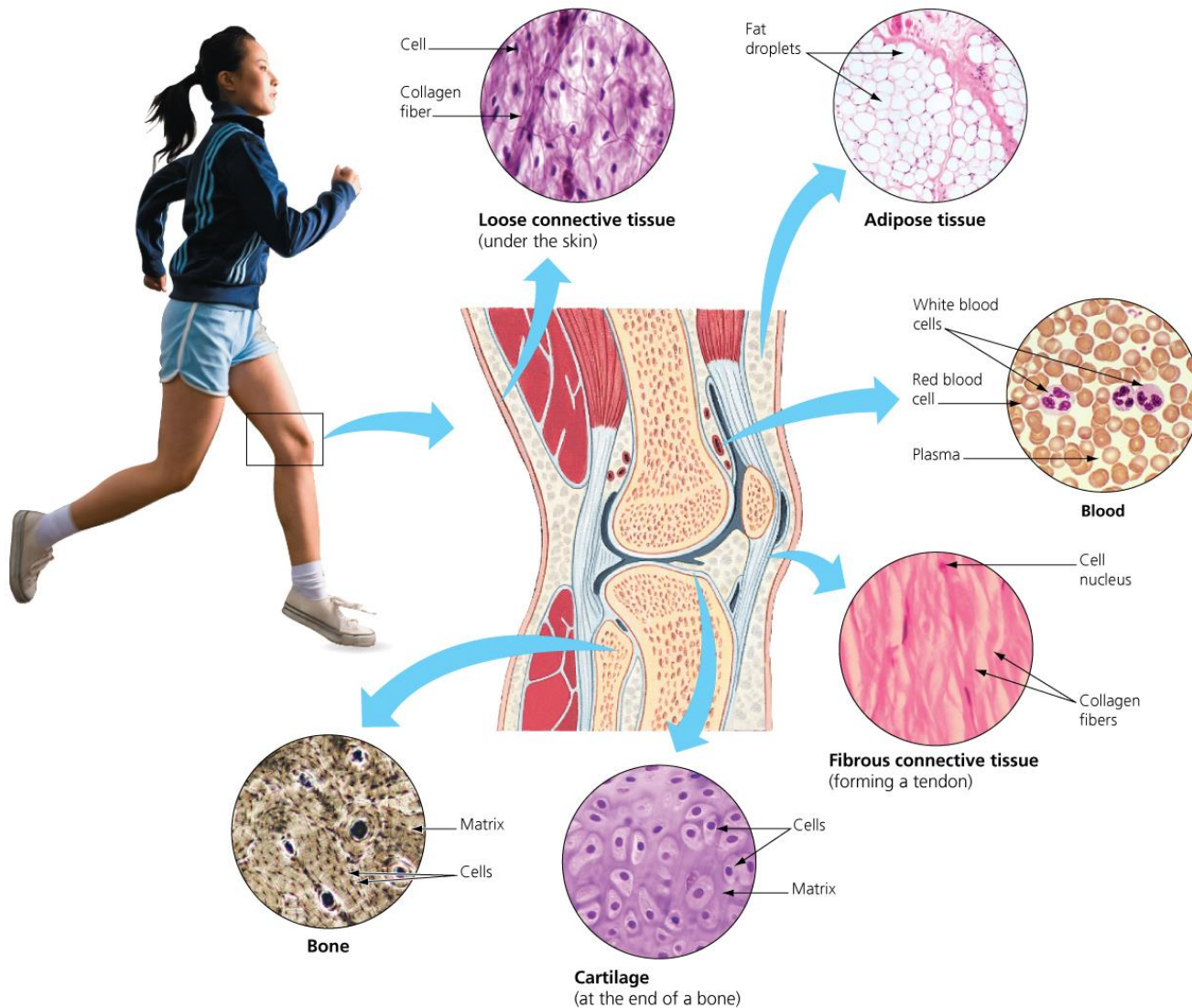
Heart  
Lung  
Stomach  
Small intestine  
Large intestine  
Urinary bladder



# Connective Tissue

- Two major functions of **connective tissue** are to support and join other tissues.
  - In contrast to epithelium, with its sheets of cells, connective tissue contains **cells scattered throughout the extracellular matrix**.
  - The structure of the extracellular matrix varies, depending on the function of the type of connective tissue to which it belongs.
  - **Figure 21.4** illustrates six of the major types of connective tissue.

# Types of Connective Tissue

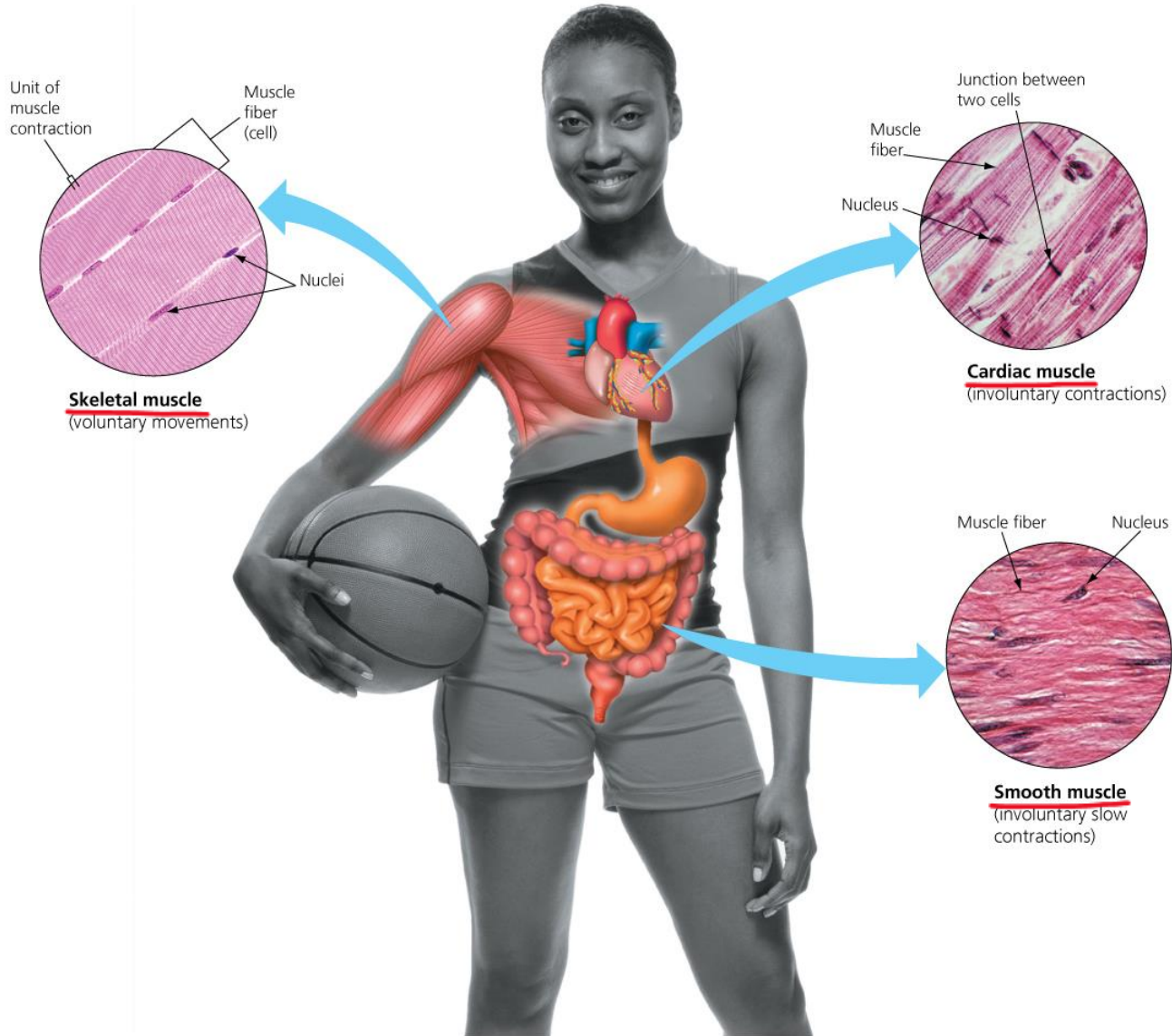


# Muscle Tissue

- **Muscle tissue** is a very abundant tissue in most vertebrates. Muscle tissue
  - consists of bundles of long, thin, cylindrical cells known as **muscle fibers** and
  - has specialized proteins arranged into a structure that contracts when stimulated by a nerve signal.
- Humans and other vertebrates have three types of muscle tissue, each with unique contractile proteins: **1) skeletal muscle, 2) cardiac (heart) muscle, and 3) smooth muscle.**



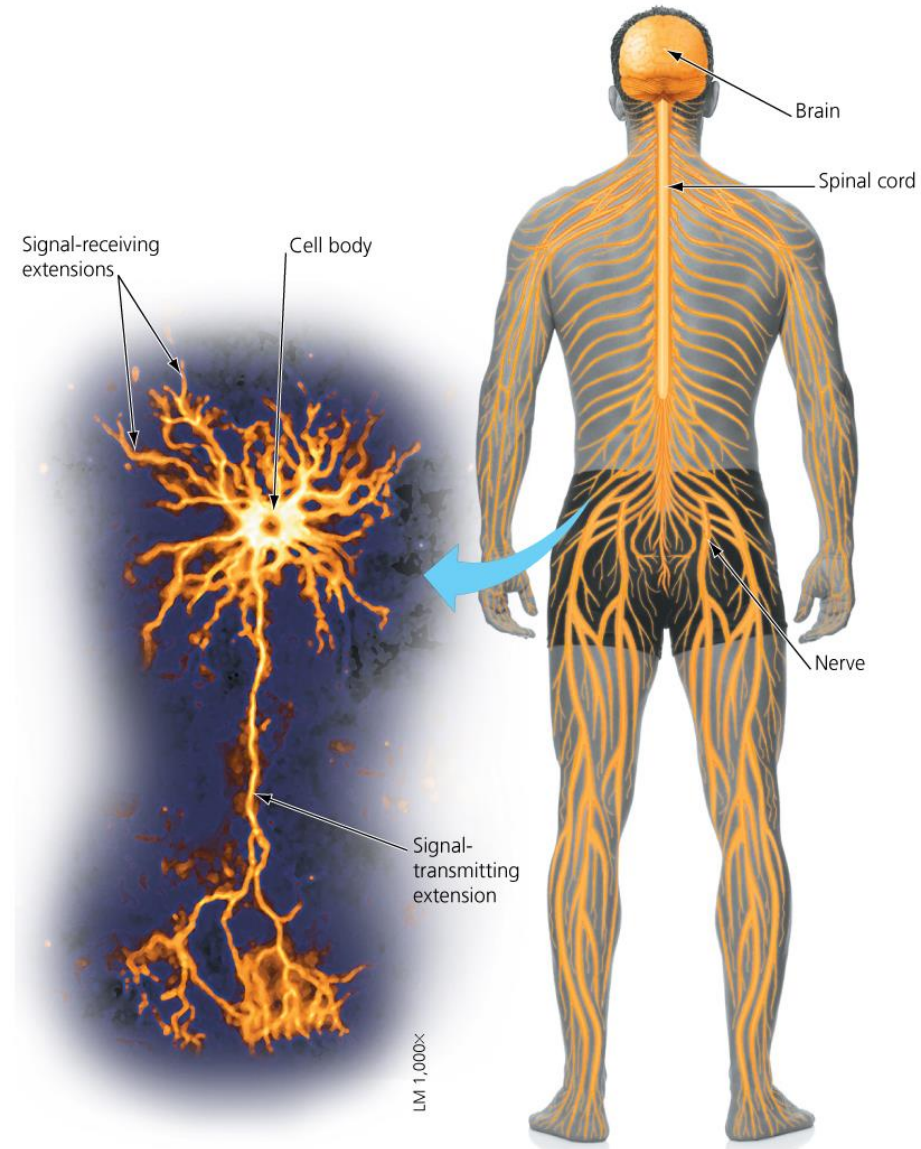
# The Three Types of Muscle Tissue



# Nervous Tissue (1 of 2)

- **Nervous tissue**
  - makes communication of information possible,
  - is found in your brain and spinal cord, and
  - consists of a network of neurons.
- The basic unit of nervous tissue is the neuron, or nerve cell.
- The transmission of information allows neurons to coordinate functions within the body.

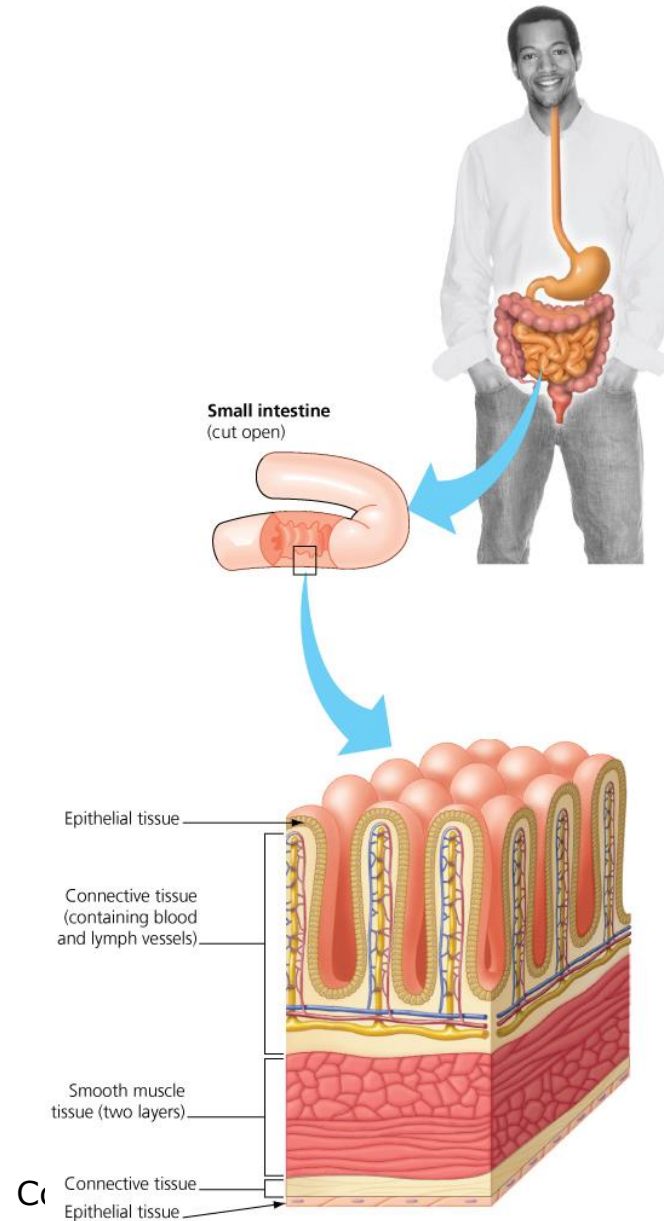
# Nervous Tissue (2 of 2)



# Organs and Organ Systems (1 of 2)

- An **organ** consists of **two or more tissues** packaged into one working unit that performs a specific function.
- To see how multiple tissues interconnect in a single organ, examine the layered arrangement of tissues in the wall of the small intestine in **Figure 21.7**.

# Tissue Layers of the Small Intestine, an Organ

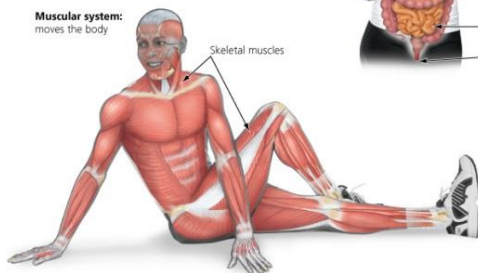
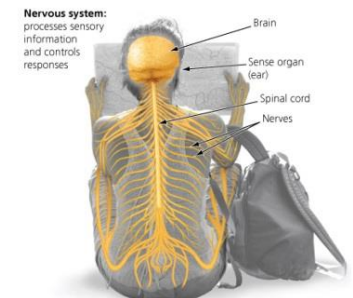
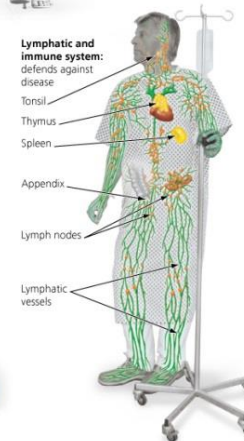
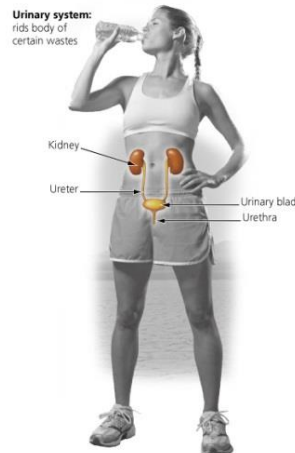
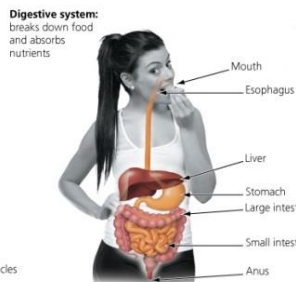
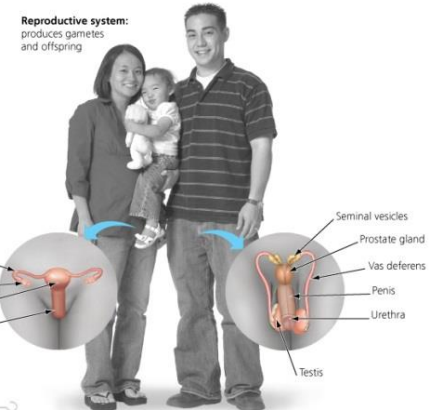
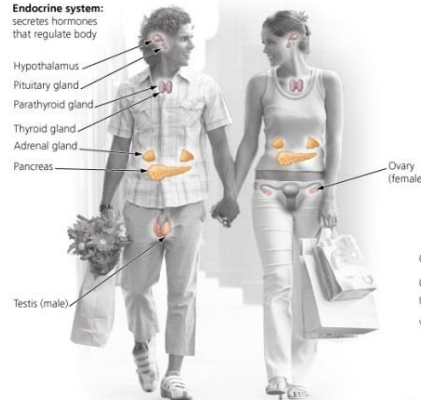
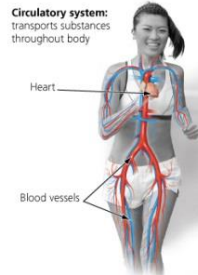
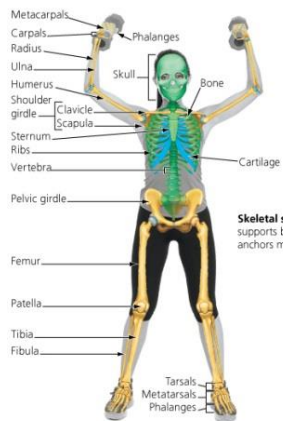


# Organs and Organ Systems (2 of 2)

- **Organ systems** are **teams of organs** that work together and perform vital body functions.
- An organism depends on the interconnection of all its organ systems for survival.
- Your body is a whole, living unit that is greater than the sum of its parts.
- **Figure 21.8** presents an overview of 11 major organ systems in vertebrates.

Organ system: 기관계

# Human Organ Systems



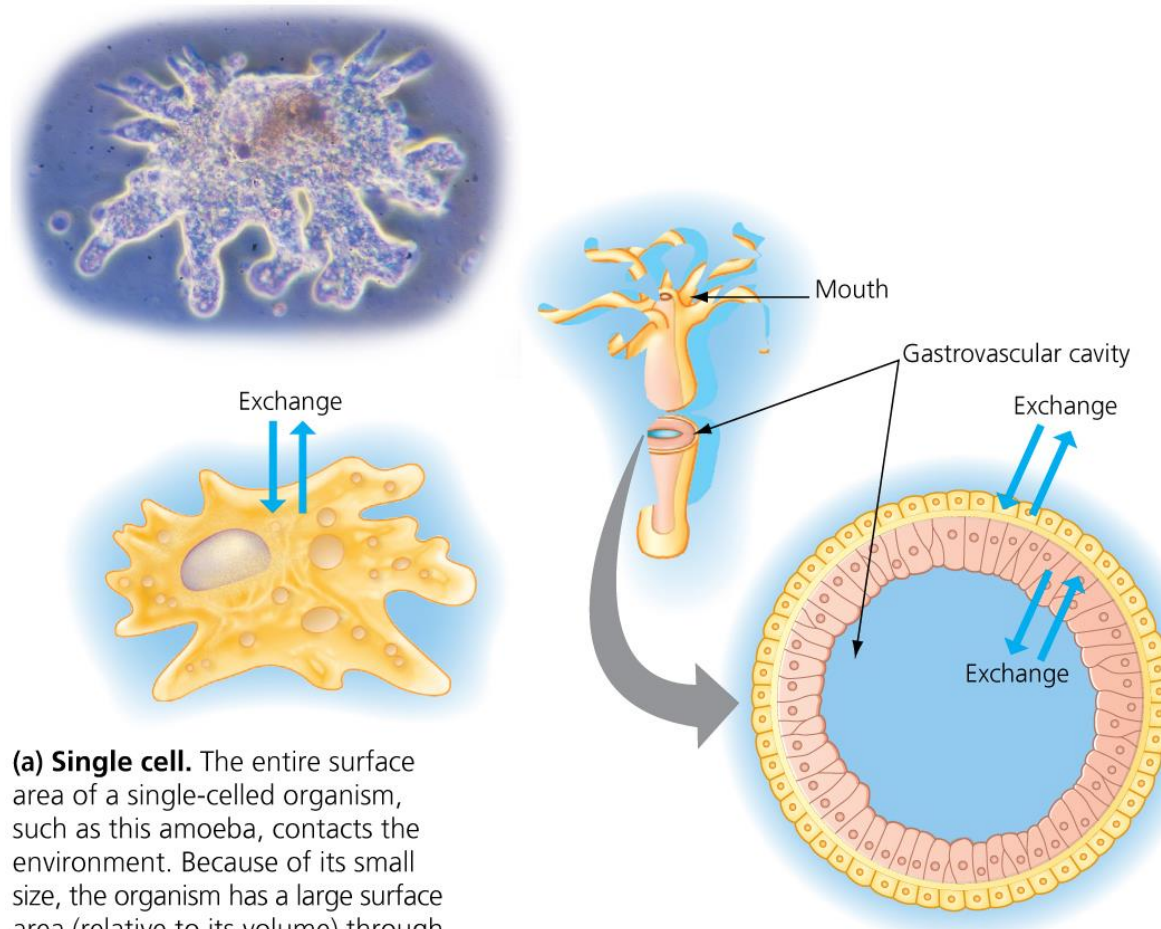


# Exchanges with the External Environment

(1 of 3)

- Every organism is an **open system** that continuously exchanges chemicals and energy with its surroundings. The ability to transform energy and matter is one of life's unifying themes.
  - An animal's size and shape affect its exchanges with its surrounding environment.
  - Every living cell of an animal's body must be bathed in a watery solution, partly because substances must be dissolved in water to cross cell membranes.
  - In a single-celled amoeba, every part of the cell's membrane touches the watery environment.

# Contact of Simple Organisms with the Environment



**(a) Single cell.** The entire surface area of a single-celled organism, such as this amoeba, contacts the environment. Because of its small size, the organism has a large surface area (relative to its volume) through which it exchanges materials with the external world.

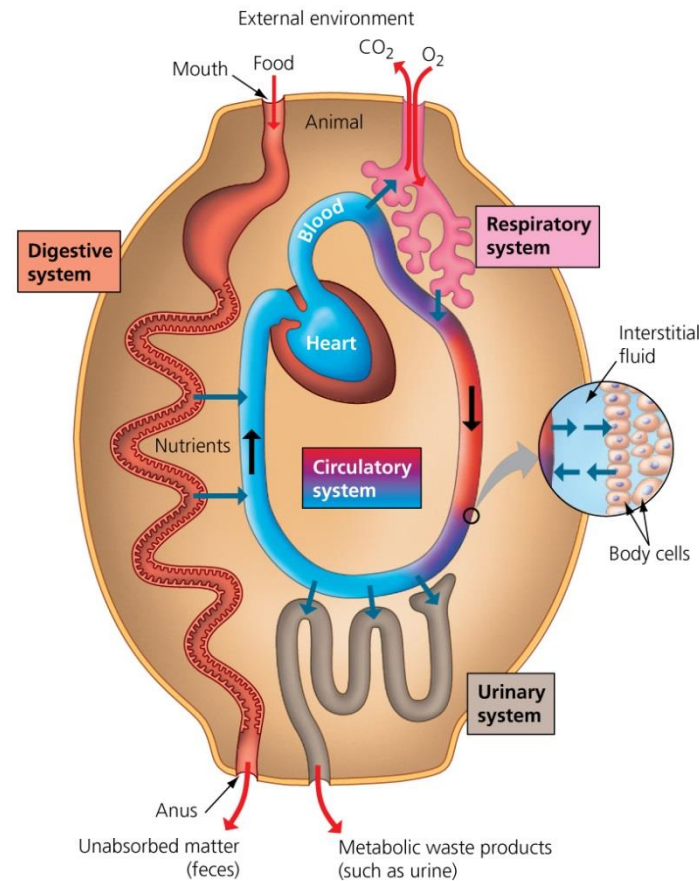
**(b) Two cell layers.** Although a hydra is multicellular, each one of its cells touches an aqueous environment. The body of the hydra has only two cell layers, both exposed to water.

# Exchanges with the External Environment

(2 of 3)

- Exchange with the environment is more complicated for **complex, multilayered animals**.
  - Each cell in a multicellular organism has a plasma membrane where exchange can occur.
  - But this exchange only works if all the cells of the animal have access to a suitable watery environment.
  - **Figure 21.10** shows a schematic model of an animal body, highlighting the three organ systems (digestive, respiratory, and urinary) that exchange materials with the external environment.

# Exchange Between the External Environment and the Internal Environment of Complex Animals



Interstitial: 사이의

# Exchanges with the External Environment

(3 of 3)

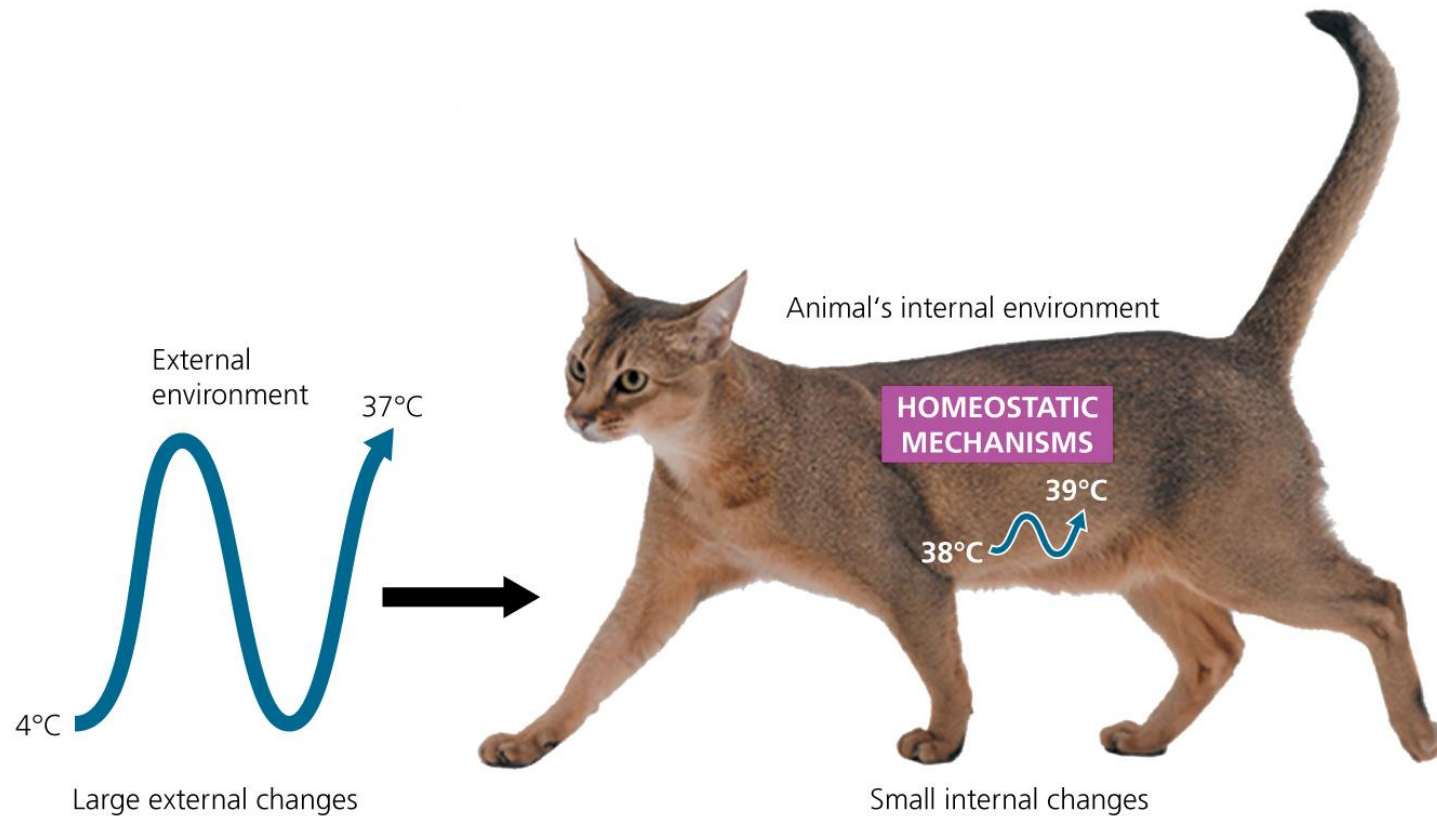
- Complex animals have evolved extensively folded or branched internal surfaces that maximize surface area for exchange with the immediate environment.
- For example, your lungs exchange oxygen and carbon dioxide with the air you breathe. Lungs are not shaped like big balloons but are in fact millions of tiny hollow structures at the tips of finely branched air tubes

# Regulating the Internal Environment:

## Homeostasis

- Animals adjust to a changing environment.
- The internal environment of vertebrates includes the **interstitial fluid** that fills the spaces between cells and exchanges nutrients and wastes with microscopic blood vessels.
- **Homeostasis**, which literally means “steady state,” is the tendency to maintain relatively constant conditions in the internal environment even when the external environment changes.

# An Example of Homeostasis

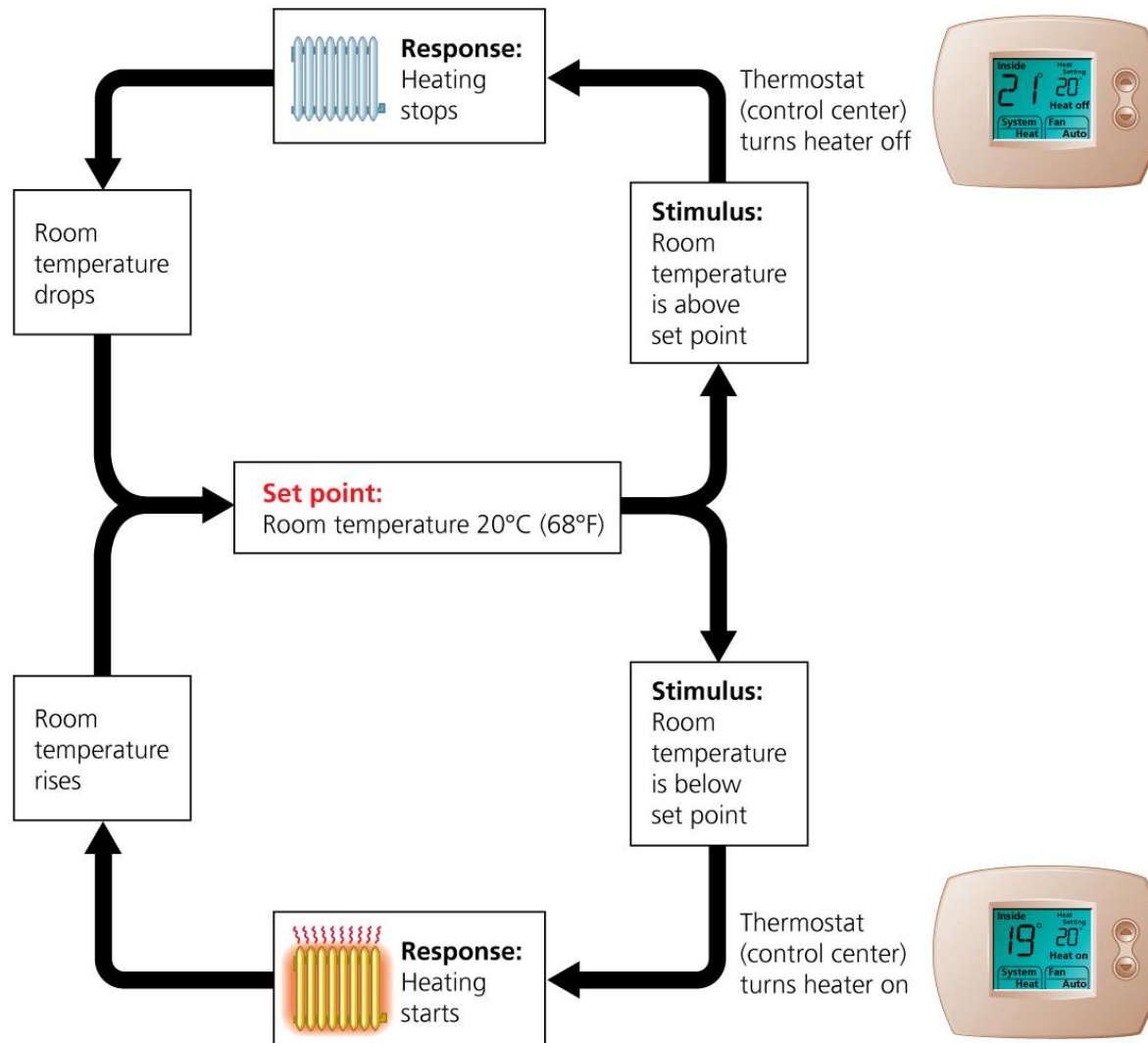




# Negative and Positive Feedback

- Most mechanisms of homeostasis depend on a principle called **negative feedback**, in which **the results of a process inhibit that same process**, such as a thermostat that monitors temperature and switches the heater on and off.
- Less commonly, organisms use **positive feedback**, in which the results of a process **intensify** that same process.

# An Example of Negative Feedback: Control of Room Temperature



# Thermoregulation (1 of 3)

- The homeostatic mechanism that controls temperature is called **thermoregulation**.
  - The ability to maintain a body temperature substantially warmer than the surrounding environment is characteristic of **endotherms**, animals such as mammals and birds that derive most of their body heat from their own metabolism.
  - In contrast, **ectotherms**, which include most invertebrates, fishes, amphibians, and nonbird reptiles, obtain their body heat primarily by absorbing it from their surroundings.

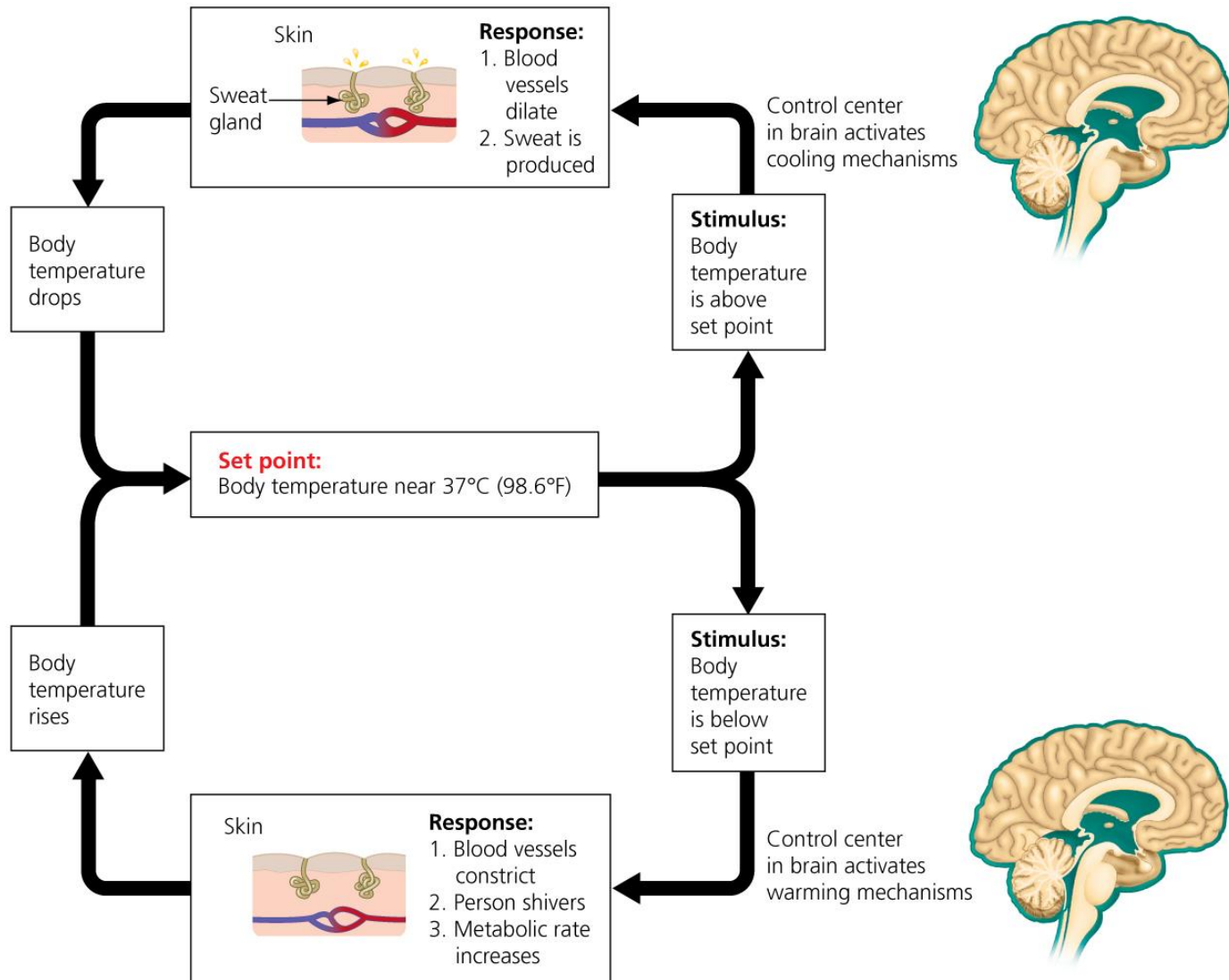
Endotherm: 온혈 동물

Ectotherm: 변온 동물, 냉혈 동물

# Thermoregulation (2 of 3)

- You have a number of structures and mechanisms that aid in thermoregulation.
- As shown in **Figure 21.14**, your brain has a control center that maintains your body temperature near 37°C (98.6°F).

# Thermoregulation in the Human Body



# Thermoregulation (3 of 3)

- **Fever**, an abnormally high internal temperature, is a body-wide response that usually indicates an ongoing fight against infection.
  - Many people mistakenly believe that the invading microbes themselves cause a fever. In fact, the cause is usually the body's fight *against* the microbes.
  - A moderate fever of 38–39°C (100–102°F), however, discourages bacterial growth and speeds the body's internal defenses.

# The Process of Science: How Does a Python Warm her Eggs? (1 of 2)

- **Background:**

- Ectotherms usually absorb heat from their surroundings. However, some ectotherms can generate their own body heat.
- Researchers at the Bronx Zoo in New York City noticed that a female Burmese python incubating her eggs wraps her body around them and frequently contracts the muscles in her coils.
- The researchers wondered if the snake's muscle contractions serve to warm the eggs during incubation, similar to the way that shivering can warm your body.



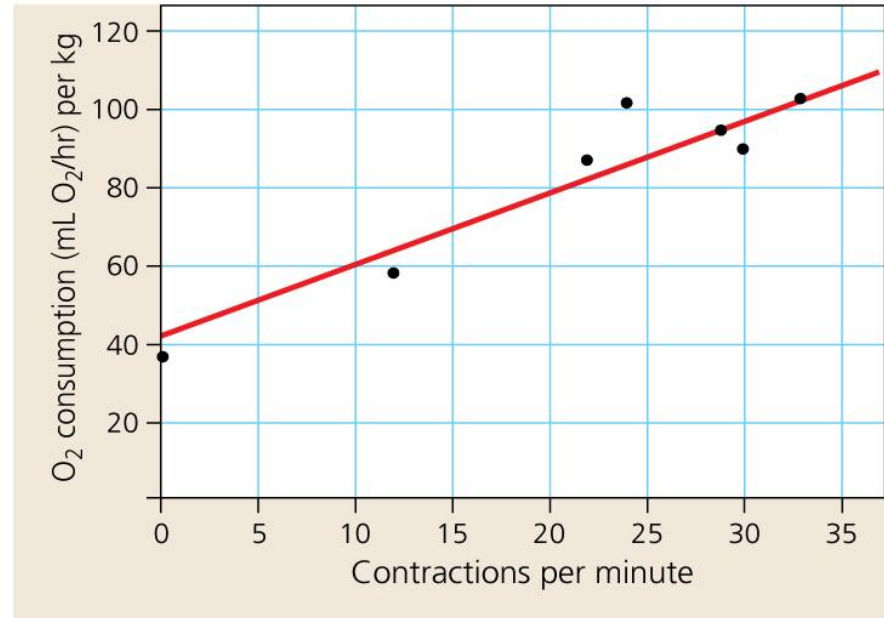
# The Process of Science: How Does a Python Warm her Eggs? (2 of 2)

- **Method:**
  - Researchers placed a female Burmese python and her eggs in a chamber and **varied the chamber's temperature**.
  - They monitored the rate of the snake's **muscle contractions** and her **oxygen uptake**, a measure of **the rate of cellular respiration**.
- **Results:** The python's oxygen consumption increased when the temperature in the chamber decreased.

# Measuring Heat Production by a Burmese Python



(a) A Burmese python incubating a clutch of eggs.



(b) Oxygen consumed by a Burmese python as it contracts its muscles.

# Osmoregulation (1 of 2)

- Living cells depend on a precise balance of water and solutes.
- **Osmoregulation** is the control of the gain or loss of water and dissolved solutes.
  - Osmoregulation is based largely on regulating solutes because water follows the movement of solutes by osmosis.
  - Osmosis occurs whenever two solutions separated by a cell's permeable membrane differ in their total solute concentrations. There is always **a net movement of water** from the solution with lower solute concentration to the one with higher solute concentration.

# Osmoregulation (2 of 2)

- **Saltwater fish** lose water by osmosis because there is less salt in their tissues than the water in which they swim.
- **Freshwater fish** have the opposite problem: The external solute concentration is low, so water enters the fish by osmosis.
- Most land animals lose water through urinating, defecating, breathing, and perspiring but can counterbalance the loss by eating and drinking.

# Counterbalancing water loss through drinking.



# Homeostasis in the Urinary System (1 of 7)

- The urinary system plays a central role in **osmoregulation**, regulating the amount of water and solutes in body fluids by retaining water when we are dehydrated and expelling it when we are hydrated.
- Besides osmoregulation, the urinary system plays another important role—the excretion of wastes.
  - Chief among the waste products to be excreted in urine is **urea, a nitrogen-containing compound** produced from the breakdown of proteins and nucleic acids.

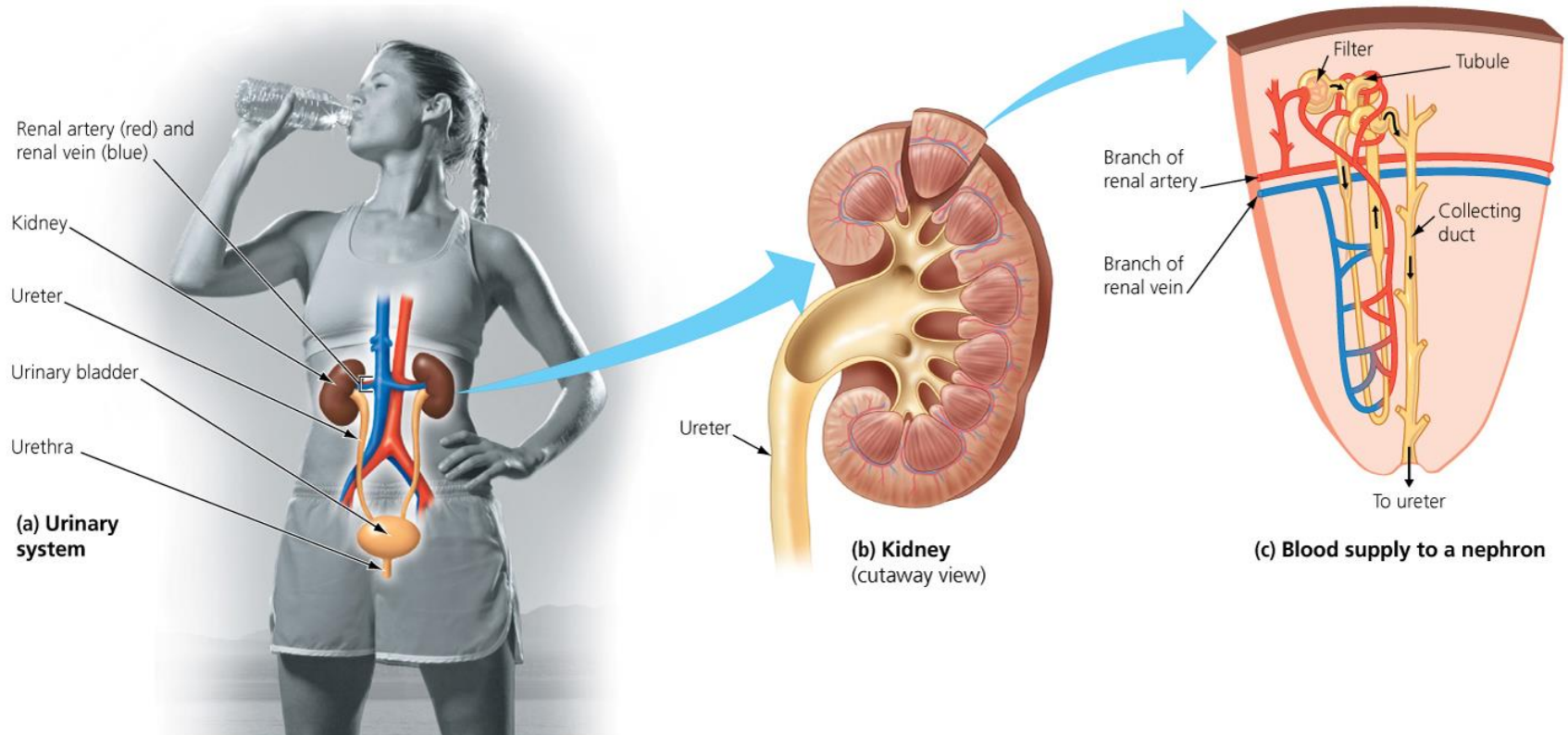
# Homeostasis in the Urinary System (2 of 7)

- In humans, the two **kidneys** are the main processing centers with nearly 100 miles of thin **tubules** and an intricate network of capillaries.
  - As blood circulates through the kidneys, a fraction of it is filtered and plasma enters the kidney tubules, forming **filtrate**.
  - Filtrate contains valuable substances that need to be reclaimed (such as water and glucose) and substances to be eliminated, such as urea.
  - The anatomy of the human urinary system is shown in **Figure 21.17**.

Filtrate: 여과된 액체



# Anatomy of the Human Urinary System





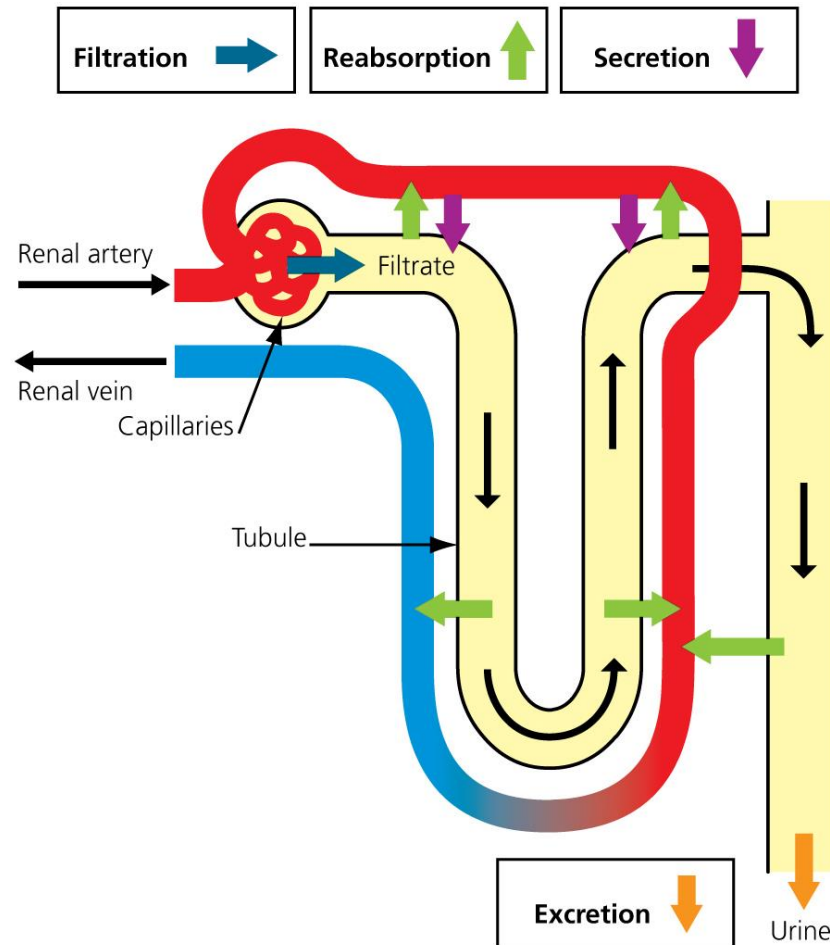
# Homeostasis in the Urinary System (3 of 7)

- **Nephrons**
  - carry out the functions of the urinary system,
  - consist of a tubule and its associated blood vessels, and
  - number more than a million in a kidney.

# Homeostasis in the Urinary System (4 of 7)

- **Nephrons** perform four key functions:
  1. **Filtration** occurs as water and other small molecules are forced out of the blood when it passes through capillary walls into the kidney tubule, forming filtrate.
  2. **Reabsorption** reclaims water and valuable solutes from the filtrate and returns them to the blood.
  3. **Secretion** of certain substances, such as some ions and drugs, is transported into the filtrate.
  4. **Excretion** of urine from the kidneys to the outside of the body takes place.

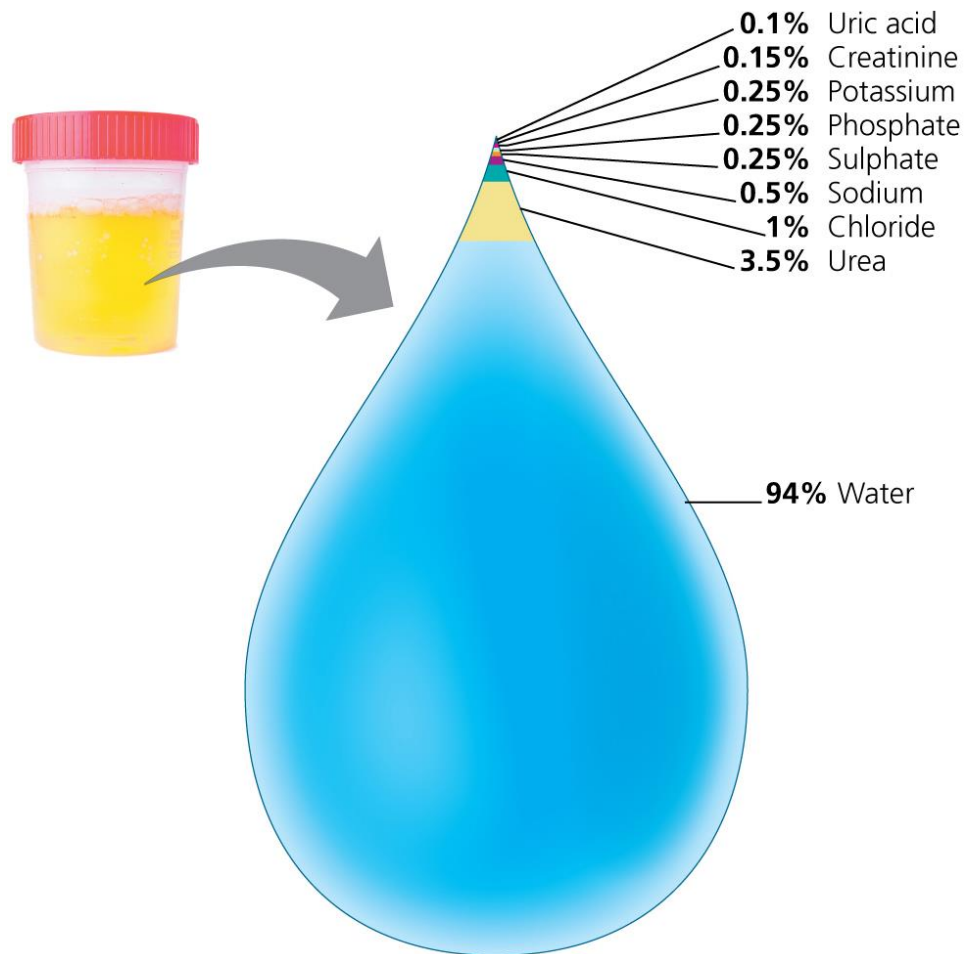
# Major Functions of a Nephron



# Homeostasis in the Urinary System (5 of 7)

- The vast majority of urine is water, with urea being the primary solute.

# The Composition of Human Urine



# Homeostasis in the Urinary System (6 of 7)

- **Hormones** regulate the kidney's nephrons to maintain water balance and are central to the interconnections of the nervous, endocrine, and urinary systems.
- Understanding filtration, reabsorption, and secretion will help you see why urine samples are often used to assess health.

# Homeostasis in the Urinary System (7 of 7)

- Factors affecting urine composition include
  - the environment,
  - presence of disease,
  - secretions from the various harmless microbes that can live in the lower part of the urethra, and diet.
- Regulating the amount of water and solutes in body fluids is just one of many examples of homeostasis.

# Evolution Connection: Adaptations for Thermoregulation (1 of 3)

- Natural selection has promoted a wide variety of adaptations for thermoregulation, including anatomical, physiological, and behavioral adaptations.
- A major anatomical adaptation in mammals and birds is insulation, consisting of hair (fur), feathers, or fat layers.



# Evolution Connection: Adaptations for Thermoregulation (2 of 3)

- Some adaptations are **physiological**, such as
  - land mammals and birds reacting to cold by raising their fur or feathers,
  - hormonal changes tending to boost the metabolic rate of some birds and mammals, increasing their heat production in cold weather,
  - shivering to produce heat as a metabolic by-product of the contraction of skeletal muscles, and
  - panting and sweating to greatly increase cooling.

# Evolution Connection: Adaptations for Thermoregulation (3 of 3)

- **Behavioral** responses regulate body temperature.
  - Some birds and butterflies migrate seasonally to more suitable climates.
  - Other animals, such as desert lizards, bask in the sun when it is cold and find cool, damp areas or burrows when it is hot.
  - Emperor penguins huddle together to stay warm.
  - Many animals cool themselves by bathing.
- These mechanisms show how a single selection pressure can drive the evolution of a multitude of adaptations in a wide variety of environments.

Bask: (햇볕을) 쬐이다

# Methods of Thermoregulation in Animals

| METHODS OF THERMOREGULATION                                                        |                                                                                     |                                                                                         |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <u>Anatomical Adaptations</u><br>(such as hair, fat, and feathers)                 | <u>Physiological Adaptations</u><br>(such as panting, shivering, and sweating)      | <u>Behavioral Adaptations</u><br>(such as bathing, basking, hibernating, and migrating) |
|  |  |     |
| Seals have a <u>thick layer of fat</u> for insulation.                             | Sheep <u>pant</u> to cool themselves.                                               | Elephants spray themselves with <u>water to cool off</u> .                              |

# Copyright



**This work is provided solely for the use of instructors in teaching their courses and assessing student learning. Dissemination or sale of any part of this work (including on the World Wide Web) will destroy the integrity of the work and is not permitted. The work and materials from it should never be made available to students except by instructors using the accompanying text in their classes. All recipients of this work are expected to abide by these restrictions and to honor the intended pedagogical purposes and the needs of other instructors who rely on these materials.**